Module 4: Evaluation

Evaluating Expressions

Values, values, and more values!

0.

Assignment Review

Parser Final

Parser Final (WIP)

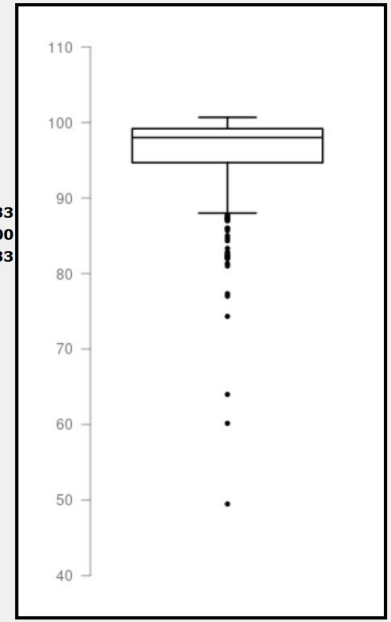
Average 95.2%, Median 98.0%

Dangerously better than expected…

- Error handling is a recurring factor.
- *Many* submissions allowing types?
- Most missed test cases:
  - Integer Exponent (Bonus)
  - Character/String escapes
  - ObjectExpr Name + LET/DEF order
  - Return value/IF interactions

Population size: 372
Median: 98
Minimum: 49.5
Maximum: 100.67
First quartile: 94.67
Third quartile: 99.17
Interquartile Range: 4.5
Outliers: 49.50 60.17 64.00 74.33 77.00 77.33 81.00 81.33 82.00 82.00 82.33 82.33 82.33 82.67 82.83 83.33 84.33 84.67 85.00 85.67 86.00 86.00 87.00 87.00 87.33 87.33 87.33 87.67 87.67

Mean
Geometric mean
Harmonic mean
Median
Mode
Standard Deviation
Variance
Mean absolute deviation
Range
Interquartile range
Quartiles
All dispersion data



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Expressions

AKA Appearance vs Personality

Expressions

Definitions

Expression referrers to code that (does, may*) return a **value**.

- Some only return values:
  - `1, 2 + 3, variable`
- Others may also have side effects!
  - `print("Hello, World!")`
  - `object.property` (!)
    - Syntax sugar for getProperty

Evaluation Order

Recall **Evaluation Order** determines the order that code is evaluated in.

- Most operators are *left associative*:
  - 1 + 2 + 3 == (((1) + 2) + 3)
- Assignment is notably *right associative*:
  - x = y = z == (x = (y = (z)))
- Evaluation order matters most when dealing with side effects (e.g. mutation) or errors!
  - `x = y = z`: If x is undefined should z be evaluated? Should y be assigned?
  - `x + f(1)`: If x is undefined should f be evaluated? What if x is the wrong type?

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